STAI100 · INTRODUCTION TO AGENTIC AI · FINAL CAPSTONE · GROUP 04

E.Z.R.A.

A Faculty Onboarding Concierge

An agentic RAG assistant that answers Faculty Manual questions with page-level citations, and verifies onboarding documents through a computer-vision pipeline.

Powered by Google Gemini 3.1 Flash-Lite (chat / agent)
Gemini 2.5 Flash (multimodal document extraction)

TEAM · STAI100 G04

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THE PROBLEM

Two obligations a new hire cannot quickly meet alone

Every newly hired DLSU faculty member arrives owing the institution atleast two things before they can teach, and today both are handled by a person, through emails and shared files.

1 Read the Faculty Manual

A **203-page** governing document, hiring criteria, teaching load, grade deadlines, dress code, sanctions.

2 Submit the paperwork

Faculty-specific documents (original TOR, three references, physical-fitness certification, prior-employer clearance) **plus national statutory ones**

WHO FEELS IT

The new faculty member
waiting days for an answer to a policy question they cannot look up themselves.

Chairs, HR & Faculty Affairs
re-answering the same onboarding questions for every incoming cohort, often inconsistently.

The institution
carrying repeated, manual effort that scales with every hiring term.

BUSINESS CASE

Quantifying the current manual process

A conservative, unit-disciplined estimate of recurring effort at one representative institution (DLSU).

1,432
faculty at DLSU
585 full-time + 847 part-time
(De La Salle University, n.d.)

›

× 5%
turnover / renewal
Per term, conservative assumption, below the figure based on US-based CUHA Report (2024)

›

72
new / renewing hires
per term (lower-bound assumption based on the conservation assumption)

›

× 5
questions each
hypothetically stating, if each new/renewing hire asks at least 5 question

›

360
policy questions
per term

360 questions × 21 min (midpoint of the figure ranges across independent sources)

~ 126 staff-hours

spent every term answering the same questions again and again.

AND, ON THE DOCUMENT SIDE

~ 3 min of manual “eyeballing” per document

A deterministic validation layer eliminates that cost per document, rather than merely shortening it. (Sources converge on a 1-5 min per-document range.)

Figures are unit-disciplined estimates from independent HR-analytics sources; hire counts are a bounded assumption, not a sourced DLSU figure. Full reference list can be seen at the end.

WHY NOT JUST CHATGPT?

The difference of E.Z.R.A

1

No institutional grounding

A general model cannot tell a real Manual provision from a plausible guess about how university policy “usually” works. The two read identically to the person asking.

2

Verification needs state & rules

Extraction is the easy half. Validating fields against per-hire records and an employer-specific freshness window is deterministic business logic a chatbot cannot supply.

The agentic answer: retrieve the governing text, cite the page, decline when the evidence isn’t there, and gate every document decision on deterministic rules, none of which a single prompt can guarantee.

PROPOSED SOLUTION

One agent, two tracks, one conversation

PRIMARY

Faculty Manual Q&A

- RAG over the real 203-page DLSU Faculty Manual 2021
- Grounded answers with page-level citations, hard-checked, never invented
- Asks which faculty class you are when the answer differs
- Says "I don't know" + routes to a human when the Manual is silent

SECONDARY

NBI Clearance + ID Verification

- Computer-vision track, the mandatory CV/DS component
- OpenCV quality gate → Gemini extraction → deterministic rules
- Checks completeness, correctness & 6-month freshness, not forgery
- Accept / reject / needs-human-review, model never has final say
- Both verified →HR receives an auto-composed handoff packet

TARGET USERS & STAKEHOLDERS

Whose need each capability serves

New faculty member

PRIMARY USER

Need

Needs a fast, trustworthy answer and a clear submission checklist, without waiting days on email.

How we serve it

Immediate cited answers; a live "what have I submitted?" checklist.

Faculty Affairs

PRIMARY BENEFICIARY

Need

Absorbs repeated onboarding questions and manual document checks every term, often inconsistently.

Deflects routine questions; auto-validates documents with a human-review fallback.

The institution (DLSU)

SPONSOR

Need

Wants consistent policy answers and reclaimed staff time that scales with hiring.

How we serve it

Consistent grounding across departments; recurring effort removed per term.

Other PH universities

FUTURE ADOPTER

Need

Face the same manual + statutory-document burden with their own handbook.

How we serve it

Retarget by swapping corpus + config, no retrieval / validation / UI rewrite.

SYSTEM OVERVIEW

The modules behind one chat window

1 Intent Router

Classifies each message: Manual question, document upload/status, ambiguous, or out-of-scope.

2 RAG Retrieval

Finds the right passages in the Manual (dense + BM25 hybrid) and grounds every answer.

3 Web Search Tool

Domain-restricted .gov.ph fallback for statutory how-to questions the Manual can't answer.

4 CV Pipeline

OpenCV quality gate + Gemini multimodal extraction for uploaded NBI clearances and government IDs, automatic HR handoff.

5 Validation Rules

Six deterministic checks decide accept / reject / needs-review, not the model.

6 Guardrails

Injection, toxicity, PII and citation-grounding checks on every turn.

7 Memory

Session history, remembered faculty class, and a per-hire onboarding checklist.

8 Monitoring

MLflow traces: latency, tokens, citation counts, validation outcomes, sanitized.

KEY FEATURES

Every capability traces to a need and a value

NEED / RISK	SYSTEM CAPABILITY	USER BENEFIT	BUSINESS VALUE
Confident wrong answers about employment terms	**Grounded RAG with page-level citations, hard-verified against retrieved chunks**	Trust the answer, and check the exact page	**Fewer costly policy errors; consistent answers across departments**
Same question has different answers by faculty class	**Faculty-class awareness, surfaces the correct per-class section rather than blending.**	No wrong-class answers; remembered for later	**Avoids mis-advising on leave, load, benefits**
Model guessing when the Manual is silent	**Abstention: “I don’t know” + routing above a similarity floor**	Honest handoff instead of a plausible guess	**Protects institution from fabricated policy**

KEY FEATURES

Every capability traces to a need and a value

NEED / RISK	SYSTEM CAPABILITY	USER BENEFIT	BUSINESS VALUE
Manual document eyeballing, one scan at a time	**3-layer document check on NBI + government ID: quality gate → extraction → deterministic rules**	Instant accept / reject / review with a reason	**~3 min saved per document; 0 bad auto-accepts**
“What have I already submitted?”	**Per-hire onboarding checklist in memory**	A live status view of what’s done and missing	**Less back-and-forth; smoother onboarding**

END-TO-END USER FLOW

One continuous onboarding conversation

1 Ask

"What do I need to submit before I start?" → cites the pre-employment checklist (faculty-specific + statutory).

2 Clarify

"How much service leave do I get?" → agent resolves the faculty class, answers the right section, and remembers it.

3 Cite

"When is my syllabus due?" → Appendix F #23: within the first two weeks, with the attached sanction.

4 Abstain

Asks a Student-Handbook grading rule → "I don't know" + a routing suggestion, instead of guessing.

5 Verify

Uploads an NBI clearance → quality gate passes → fields extracted → rules confirm name + 6-month freshness → accepted.

6 Reject early

Uploads a blurry photo → quality gate rejects it before any API call, and asks for a retake with the reason.

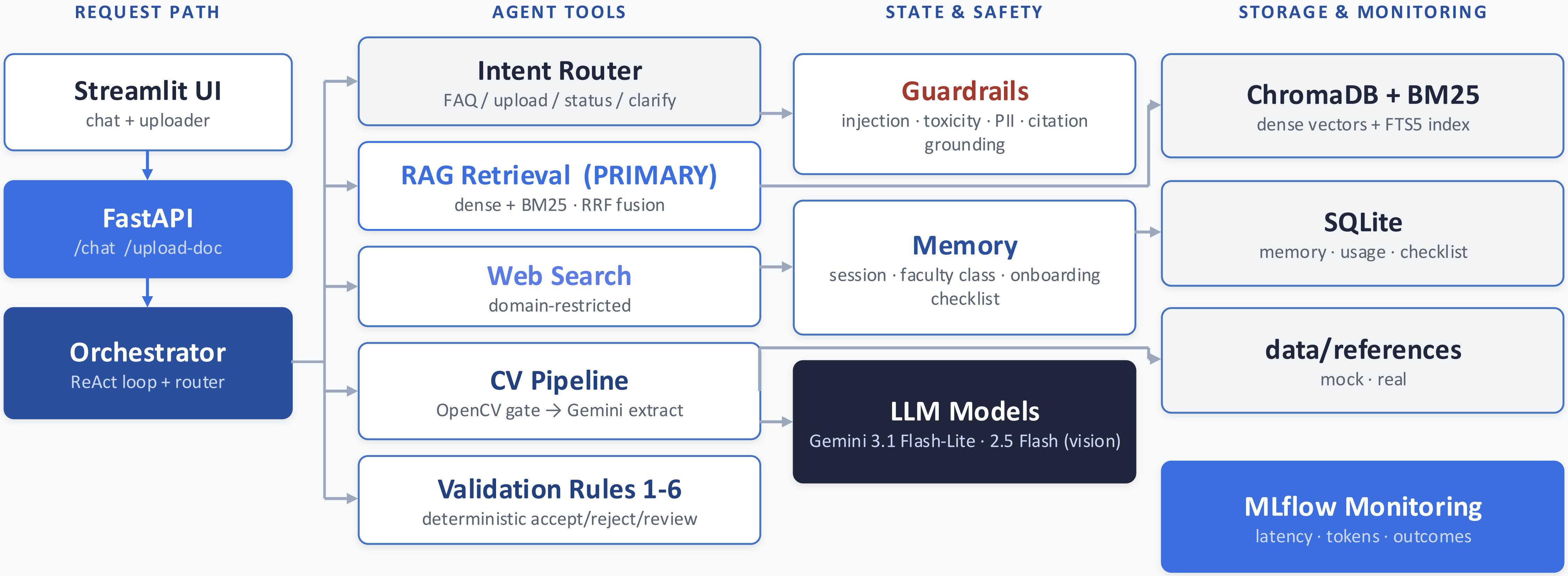
7 Track

"What have I submitted so far?" → reads the onboarding checklist state.

Every step maps to a named module, routing, RAG, memory, CV, validation, monitoring.

SYSTEM ARCHITECTURE

The High-level View of the Implementation



The CV model sits on the secondary track with the weakest authority: it proposes fields, deterministic code decides. Raw image bytes go through /upload-doc, never through the ReAct tool loop.

HOW THE CORE SYSTEM WORKS

Two pipelines

RAG

1. **Retrieve**
   dense pulls candidate chunks; a similarity floor (0.55) filters weak matches
2. **Ground**
   one answer is synthesized only from the retrieved excerpts, never model memory
3. **Verify citations**
   output guardrail hard-checks every cited page maps to a chunk retrieved this turn
4. **Abstain**
   if evidence doesn't support it → insufficient-context → "I don't know" + routing

CV

1. **OpenCV quality gate** *no LLM*
   deterministic blur / skew / exposure / resolution check, rejects bad photos before any API call
2. **Gemini extraction** *weakest authority*
   one multimodal call → typed fields, each with the verbatim text it claims to have seen
3. **Deterministic validation** *no LLM*
   six rules: type, completeness, format, identity, freshness, fail-safe → accept / reject / needs-review

DEPLOYMENT ARCHITECTURE

One Docker Compose stack, four services, one shared volume

Docker Compose
orchestrates all four services

Ingest
scripts/ingest.py · builds the index

API
FastAPI · /chat /upload-doc

UI
Streamlit · chat + uploader

MLflow
traces · latency, tokens, outcomes

shared /app/data
Chroma index · SQLite state · data/references — one volume, one consistent view

Docker Compose coordinates four services around a shared persistent volume, keeping application state, vector indexes, and reference data consistent across the stack.

REVIEW OF RELATED LITERATURE

Choosing the CV model for document extraction

MULTIMODAL LLMs COMPARED

Model	Img	JSON	Tool	Fit
OpenAI GPT-4.1	Yes	Yes	Yes	High
Gemini 2.5 Flash	Yes	Yes	Yes	High
Anthropic Claude	Yes	Yes	Yes	High
DeepSeek V4	No	Yes	Yes	Lower

Also ruled out: Tesseract & Cloud Vision need a separate field-parsing layer; Document AI needs an extra processor + config for only 1-2 doc types.

WHAT THE TASK NEEDS

A phone photo or scan must be judged usable, have its required fields extracted, and be validated to accept / reject / needs-review, so the model needs image input, structured (schema) output, and tool calling in a single call.

Chosen: Gemini 2.5 Flash

Multimodal image input, structured output and tool calling in one call, and it reuses the project's existing Google GenAI SDK + Pydantic schemas, so the CV and chat tracks share one provider. The extractor returns the downstream validation shape directly, with no free-text parsing.

REVIEW OF RELATED LITERATURE

Choosing the retrieval approach, and how it scored

RETRIEVAL-SIDE EXPERIMENT

Setup 1: Dense: semantic vector search

1. Question is turned to 768-number vector (embedding) by Gemini.
2. Every chunk in the Faculty Manual was already turned into a vector at ingest time.
3. Chroma finds the chunks whose vectors are closest to the question's vector (cosine similarity) and returns the top-k.

Setup 2: Hybrid (Dense + BM25): merges Dense and BM25

1. Dense – exactly as before
2. Best Matching 25 (BM25) – matches on **literal words**
3. **RRF fusion (Reciprocal Rank Fusion)** — combines the two ranked lists into one.

Chosen: Dense

Deliberately rejected for speed

- Query rewriting, adds a Gemini call per question
- LLM reranking, adds another call per question

RAG RETRIEVAL RESULTS

32-question golden set

100%
hit-rate @ k=8

0.90
mean reciprocal rank

100%
abstention (9 / 9)

Metric	Dense	Hybrid
Hit-rate @ 1	82.6%	82.6%
Hit-rate @ 3	95.7%	95.7%
Hit-rate @ 5	100%	95.7%
Hit-rate @ 8	100%	100%
MRR	0.90	0.89

Both reach 100% recall by k=8. Hybrid lifts the exact-term / disambiguation tier (hit@3 0.80 → 1.00), trading slightly on lookup, which is why it ships as a selectable mode.

n = 23 answerable (with an expected source) + 9 negative.

AGENT EVALUATIONS

Guardrail Evaluation

100%	100%	100%	100%
Injection/toxicity block-rate	PII detection rate	Off-topic block-rate	Jailbreak block-rate

EVALUATION & OBSERVED RESULTS

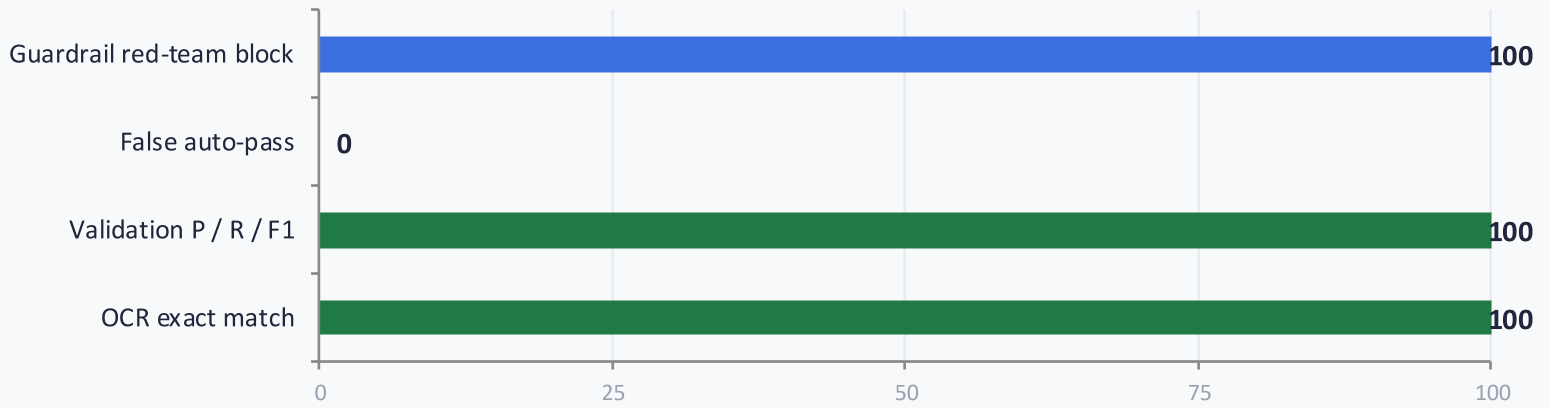
Partial Agent Evaluations

RAG Correct Hit Evaluation

- **Hit-rate @ 8 = 100%**
  every answerable question retrieved by k=8
- **MRR = 0.90**
  the correct chunk ranks near the top
- **Abstention = 100%**
  9 of 9 negatives correctly declined
- **Dense vs hybrid**
  both 100% @ k=8

CV Implementation Evaluation

Full-scale run · paid key · Aug 10, 2026



0%
false auto-pass
both doc types

100%
OCR fields exact
all 8 identities + 2 doc

100 to 75%
date_printed w/o
preprocessing

27.3%
cross-doc mismatch
0% false-trigger

AGENT EVALUATIONS

End to End Evaluation

ANSWER-LEVEL EVAL · GEMINI-AS-JUDGE

4.96	4.85	0.90	1.00
PASS · BAR ≥ 4.0	PASS · BAR ≥ 4.0		
Mean faithfulness (1–5)	Mean answer correctness (1–5)	context precision	context recall

BUSINESS VALUE

What the reclaimed effort is worth

AT ONE INSTITUTION,

126

staff-hours of repeat Q&A per term

+ ~3 min eliminated per document

verified, not just shortened

Reclaimed staff time

~120 h/term of Chair / HR / Faculty-Affairs effort redirected from repeat lookups to higher-value work, after a conservative 95% deflection assumption.

Cost anchor (reference)

Valued against the PH Salary Standardization Law SG 22-24 hourly rate, a real government reference point, not DLSU's private pay.

Consistency & correctness

One grounded source of policy answers across departments, fewer confidently-wrong replies about employment terms.

Generalizability

Retarget to another PH university by swapping corpus + config, value multiplies without re-engineering.

LIMITATIONS & FUTURE WORK

What we scoped out, and what comes next

Known limitations

- Real NBI subset (3 specimens) not yet hand-labeled, reported as live-verification, not a quantitative score
- CV track covers two document types (NBI Clearance + government ID), reported separately, never pooled
- Not authenticity / forgery detection, a completeness & correctness check by design

Future improvements

- **Complete quantitative real-subset labeling** , hand-transcribe ground truth so real docs get scored, not just live-verified
- **Extend the doc registry** , BIR 2316 and more pre-employment documents, same extract-then-validate pattern, no redesign
- **Ingest the Student Handbook** , unlock genuine multi-hop grading questions across two sources
- **Deploy at a second university** , prove the data-swap generalizability claim in practice

TEAM CONTRIBUTIONS

Divison of Tasks (STAI100 G04)

AB

Anthony Baybayon

#2 DISAMBIGUATION · #3 RAG · #9 REACT / TOOL USE · #12 ADVANCED RAG

Manual corpus & ingestion, retrieval quality, grounding & safety layers, and the tool the agent calls to reach the CV pipeline.

EAB

Ethan Axl Burayag

#4 MEMORY · #5 GUARDRAILS · #13 EVALS

The literature review & model-selection story, design tradeoffs, and full-system evaluation runs.

EDR

Ezra Del Rosario

#9 REACT / TOOL USE · #14 CV / DS DOMAIN INTEGRATION · #13 EVALS (CV)

The OpenCV quality gate, the NBI/Gemini extraction call, and the OCR / validation metrics.

MT

Mariel Tamondong

#6 SIMPLE CHAT UI · #7 API ENDPOINT DEPLOYMENT · #8 LLMOPS

The Streamlit UI, the FastAPI endpoints, MLflow monitoring, and Dockerization

Component #14 (CV/DS Domain Integration) is owned by the team as the mandatory shared component.

WHY THIS MATTERS

The working solution

A new hire's two blockers, a 203-page Manual and a hand-checked paperwork, met by **one agent** that **cites the page, admits what it doesn't know, and gates every document decision on deterministic rules a general chatbot cannot.**

Built
RAG over a real 203-page manual + a 3-layer CV verification pipeline, 12 of 14 components, plus an HR handoff built beyond the agreed scope.

Proven
100% mock-subset OCR & 0% false auto-pass; grounded, abstaining, citation-checked answers.

Worth
~126 staff-hours/term reclaimed at one institution, and a data-swap away from the next.

Now: Live Demo

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